# 1 Test Prep: Midterm 1

## 1.1

A voltage source $\vec{V} = 380\angle 25°$V at 60 Hz is connected across a system consisting of a 12Ω resistor in series with a 200 mH inductor. Compute:

a) The load current.

Finding the reactance of the inductor, $X_L = 2\pi \times 60 \times 10^{-3} = 75.40\Omega$. Therefore, the total impedance in the circuit $\vec{Z}_{\text{total}} = 12 + j75.40 = 76.35\angle 80.96°\Omega$. Then, load current is equal to

$$\vec{I} = \frac{\vec{V}}{\vec{Z}} = \frac{380\angle 25°}{76.35\angle 80.96°} = 4.98\angle{-55.96°} \text{ A.}$$

b) The load power factor.

$$\text{P.F.} = \cos\theta_{VI} = \cos(80.96°) = 0.1571 \text{ (lagging)}$$

c) The real power consumed by the load.

$$P = |V||I|\cos\theta_{VI} = 380 \times 4.98 \times 0.1571 = 297.30 \text{ W.}$$

d) The reactive power consumed by the load.

$$Q = |V||I|\sin\theta_{VI} = 380 \times 4.98 \times 0.9876 = 1.869 \text{ kVAr.}$$

## 1.2

An inductive load consists of $R_1 = 3\Omega$ in series with $X_{L1} = 4\Omega$. The load is connected parallel with another load of unknown impedance. The voltage source of the system is 120 V. The total real and reactive powers delivered by the source are 3 kW and 0 kVAr, respectively.

a) Compute $\vec{I}_1$.

First, converting the impedance of the known load to phasor form,

$$\vec{Z}_1 = \sqrt{3^2 + 4^2}\angle(\tan^{-1}\frac{4}{3}) = 5\angle 53.13°\Omega.$$

Since the voltage is the same across both reactive loads,

$$\vec{I}_1 = \frac{\vec{V}}{\vec{Z}_1} = \frac{120\angle 0°}{5\angle 53.13°} = 24\angle{-53.13°} \text{ A.}$$

b) Compute the impedance of the unknown load.

Using KCL, we know that $\vec{I}_1 + \vec{I}_2 = \vec{I}_{\text{total}} \Rightarrow \vec{I}_2 = \vec{I}_{\text{total}} - \vec{I}_1$, and given the formula for complex power, $\vec{S} = 3 \times 10^3\angle 0° = \vec{V}\vec{I}^*_{\text{total}} \Rightarrow \vec{I}^*_{\text{total}} = \frac{\vec{S}}{\vec{V}}$. (For phasors, the complex conjugate of a value is taken by changing the sign of the phase angle; magnitude is not affected).

Using the above,

$$\vec{I}_2 = \vec{I}_{\text{total}} - \vec{I}_1 = \frac{\vec{S}}{\vec{V}} - \vec{I}_1 = \frac{3 \times 10^3\angle 0°}{120\angle 0°} - 24\angle{-53.13°} = 25\angle 0° - 24\angle{-53.13°}.$$

Converting both phasors to complex form to make subtraction easier,

$$[25 - j0] - [14.4 - j19.20] = (25 - 14.4) + j(0 - 19.20) = 10.6 + j19.20 \text{ A} = \vec{I}_2.$$

Finally, converting $\vec{I}_2$ back to phasor form and finding impedance $\vec{Z}_2$,

$$\frac{\vec{V}}{\vec{I}_2} = \frac{120\angle 0°}{\sqrt{10.6^2 - 19.20^2}\angle(\tan^{-1}\frac{-19.20}{10.6})} = \frac{120\angle 0°}{21.93\angle{-61.10°}} = 5.47\angle{-61.10°}\Omega.$$

## 1.3

The current and voltage of a Y-connected load are:

$$V_{bn} = 380\angle{-90^\circ} \text{ V}$$
$$I_c = 100\angle 150^\circ \text{ A}$$

a) Compute $I_a$, $V_{ab}$, and $Z$ (the load impedance).

   For a y-connected load,

$$\begin{aligned} I_a &= |I|\angle(\theta_{I_a}) \\ I_b &= |I|\angle(\theta_{I_a} - 120^\circ) \\ I_c &= |I|\angle(\theta_{I_a} + 120^\circ), \end{aligned}$$

   so $I_a = |I_c|\angle\theta_{I_c} = 100\angle(150^\circ - 120^\circ) = 100\angle 30^\circ$ A.

   Similarly,

$$\begin{aligned} V_{ab} &= |V|\angle(\theta_{V_{ab}}) \\ V_{bc} &= |V|\angle(\theta_{V_{ab}} - 120^\circ) \\ V_{ca} &= |V|\angle(\theta_{V_{ab}} + 120^\circ), \end{aligned}$$

   and

$$\begin{aligned} V_{ab} &= |V_{an}|\sqrt{3}\angle(\theta_{V_{an}} + 30^\circ) \\ V_{bc} &= |V_{bn}|\sqrt{3}\angle(\theta_{V_{bn}} + 30^\circ) \\ V_{ca} &= |V_{cn}|\sqrt{3}\angle(\theta_{V_{cn}} + 30^\circ) \end{aligned}$$

   so $V_{bc} = |V_{bn}|\sqrt{3}\angle(\theta_{bn} + 30^\circ) = 658.18\angle{-60^\circ}$ V and $V_{ab} = |V_{bc}|\angle(\theta_{bc} + 120^\circ) = 658.18\angle 60^\circ$ V.

   Finally, $Z$ in a y-connected load is equal to

$$\vec{Z}_a = \frac{\vec{V}_{an}}{\vec{I}_a} = \frac{380\angle 30^\circ}{100\angle 30^\circ} = 3.8\angle 0^\circ\Omega,$$

   and all impedances are equal, $\vec{Z}_a = \vec{Z}_b = \vec{Z}_c$, in a balanced three-phase load.

b) Compute the power factor and the three-phase real and reactive power of the load.

   Since the phase angle of the impedance is 0°, this implies that voltage and current are in phase due to purely resistive load, so

$$P.F. = 0,$$

   neither leading nor lagging.

   Then, assuming a phase angle difference of 0°, real power is given by

$$P = 3 \times |V_{an}||I_a| = 3 \times 380 \times 100 = 114 \text{ kW}$$

   and reactive power is

$$Q = 0 \text{ VAr}.$$

## 1.4

A three-phase Y-connected source with $480\angle 0°$ V (line-to-line) is energizing a three-phase, balanced, Delta-connected load with impedance $4 + j3$ per phase. Determine:

a) The load power factor.

Finding the phase angle of the impedance, $\theta_Z = \tan^{-1}\left(\frac{3}{4}\right) = 36.87°$, then power factor is given by

$$P.F = cos(\theta_Z) = 0.8 \text{ (lagging)}.$$

b) The transmission line current $I_a$.

Transmission line current can be related to load current as

$$\vec{I_a} = \sqrt{3}|I_{ab}|\angle(\theta_{I_{ab}} - 30°),$$

and

$$\vec{I_{ab}} = \frac{V_{ab}}{Z} = \frac{480\angle 0°}{5\angle 36.87°} = 96\angle{-36.87°} \text{ A}.$$

Therefore,

$$\vec{I_a} = (\sqrt{3} \times 96)\angle(-36.87° - 30°) = 166.28\angle 66.87° \text{ A}.$$

c) The three-phase real power consumed by the load.

Using the formula given previously,

$$P = 3 \times |I_a||V_{ab}|\cos(\theta_Z) = 239.44 \times 10^3 \times \cos(0.6435) = 191.55 \text{ kW}.$$

d) The three-phase reactive power consumed by the load.

Using the same formula, but using the sine rather than the cosine of the phase angle difference,

$$Q = 239.44 \times 10^3 \times \sin(0.6435) = 143.66 \text{ kVAr}.$$